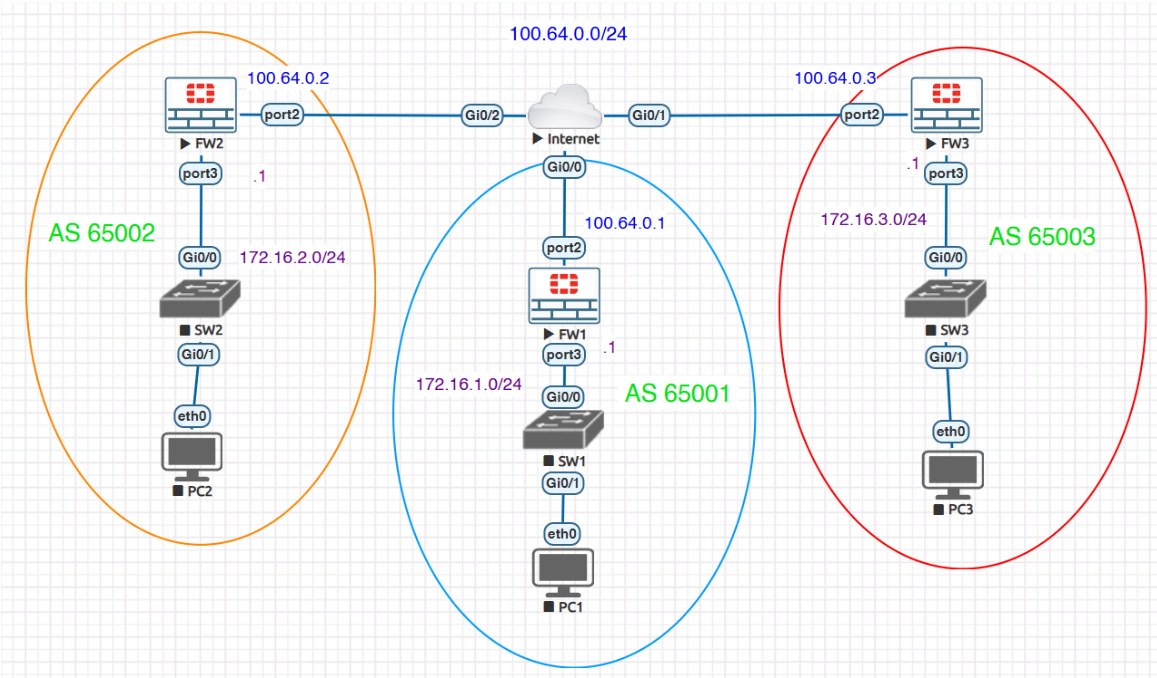
BGP with Fortigate Firewall

Objective: Configure BGP relationship between 3 fortigate firewall



Phase 1: Addressing Schema

We will use a central subnet for the "Internet" switch and unique subnets for each LAN.

Device	Interface	Role	IP Address	BGP AS
FW1	port2	WAN	100.64.0.1/24	AS 65001
FW1	port3	LAN	172.16.1.1/24	
FW2	port2	WAN	100.64.0.2/24	AS 65002

FW2	port3	LAN	172.16.2.1/24	
FW3	port2	WAN	100.64.0.3/24	AS 65003
FW3	port3	LAN	172.16.3.1/24	

Phase 2: The "Service Request" Workflow (CLI Only)

Step 1: Interface & Management Setup

You need to be able to reach these firewalls. Use the CLI to open up access on your WAN and LAN ports.

Bash

```
# Example for FW1 (Repeat logic for others)
config system interface
    edit "port2"
        set alias WAN
        set ip 100.64.0.1 255.255.255.0
        set allowaccess ping https ssh
    next
    edit "port3"
        set alias LAN
        set ip 172.16.1.1 255.255.255.0
        set allowaccess ping
    next
end
```

Step 2: Firewall Policy BGP will form neighbors, but the PCs won't be able to ping each other across the firewalls unless you have a policy.

Bash

```
config firewall policy
    edit 1
        set name "LAN_to_WAN"
```

```
        set srcintf "port3"
        set dstintf "port2"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
    next
    edit 2
        set name "WAN_to_LAN"
        set srcintf "port2"
        set dstintf "port3"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
    next
end
```

Phase 3: The BGP Configuration

We will set up **eBGP** (External BGP) so each firewall acts as its own autonomous domain.

On FW1 (The "Hub" logic):

Bash

```
config router bgp
    set as 65001
    set router-id 1.1.1.1
    config neighbor
        edit "100.64.0.2
            set remote-as 65002
        next
        edit "100.64.0.3"
            set remote-as 65003
        next
    end
    config network
        edit 1
            set prefix 172.16.1.0 255.255.255.0
        next
    end
```

(Repeat this on FW2 and FW3, changing the AS and neighbor IPs accordingly.)

Phase 4: Verification (The "No-GUI" Challenge)

Once you finish, do not open the browser. Run these commands:

1. `get router info bgp summary`: Look for **State/PfxRcd**. If it shows a number (like `1`), the neighbor is up!
2. `get router info routing-table bgp`: Do you see the `172.16.2.0` and `172.16.3.0` routes?
3. `execute ping 172.16.2.1`: Can FW1 talk to FW2's LAN? Can PC1 talk to PC2 , PC3 back and forth?